

Examiner reports

Q1.

This question was answered correctly by about half of all students. While this was a straightforward question, periodic sequences are a new topic on the specification, which may account for students' poorer performance. This question can be done very easily through the efficient use of allowed calculator functions.

The most frequently seen incorrect answers were π and 2π in roughly equal proportion; very few students selected 8.

Q2.

This question on sequences proved to be a discriminating one. The more able students were able to write down and solve two correct simultaneous equations in p and q , using the given information but other students were keen to use either arithmetic or geometric series inappropriately. A majority of students wrote down the correct equation using the information about the first and second terms, however many could not then translate the further information about the limit into a corresponding correct equation. Using the given value of p , most students answered part (b) correctly by finding the correct values for q and u_3 .

Q3.

Most candidates scored two marks out of five for this question on sequences. They were able to form the equation, ' $48 = 60p + q$ ', linking u_1 and u_2 and then used the printed value of p in order to find q . More able candidates wrote down a second equation using the information about the limit of u_n by replacing both u_{n+1} and u_n by 12 in the given formula, and then solved their simultaneous equations correctly to find the values of p and q . Sadly there are still candidates who consider, incorrectly, this topic to be a test on infinite geometric series.

In part (b), approximately 60% of the candidates scored the mark for finding the value of u_3 .

Q4.

This question on sequences was generally answered much better than similar questions in previous years. There were many correct answers to part (a), although replacing u_1 with 2 encouraged a number of weaker candidates to replace u_2 with 3. A majority of candidates did write an equation in L in part (b), although there was a significant minority who were keen to incorrectly consider use of the sum of an infinite geometric series formula instead of just replacing both u_{n+1} and u_n by L in the given formula. Some other

candidates who gave the correct equation ' $L = 6 + \frac{2}{5}L$ ' could not rearrange this equation to obtain $L = 10$.

Q5.

Most candidates were able to show that $k = 0.75$ in part (a) and the majority of these then found the correct values for u_3 and u_4 .

The most common wrong answers to part (b) were $u_3 = 32$ and $u_4 = 40$, from those who assumed that the sequence was arithmetical.

The examiners expected to see, in answers to part (c)(i), a clear indication that candidates appreciated that both $u_{n+1} = L$ and $u_n = L$ when forming the equation for the limit. Although many correct solutions were seen, there were also many other candidates whose

solutions incorrectly assumed that the sequence was geometric and the expression $\frac{a}{1-r}$ was used.

Q6.

Those candidates who formed the two equations in p and q normally went on to solve these simultaneously to find the correct values for p and q . Some candidates, however, only used one of the equations and substituted $q = 6$ to find the value of p and so never showed that $q = 6$. Many candidates found the correct value for u_4 but a significant number of candidates did not attempt part (c). Candidates who replaced both u_n and u_{n+1} by L , generally scored all the three marks in part (c). The most common wrong start to answering part (c) was to replace u_{n+1} by $L + 1$.

Q7.

The candidates' solutions for this question would seem to suggest that they were either very familiar with the topic or had not met it before. Those who used the given values for the first three terms to form two equations in p and q almost always went on to solve the simultaneous equations correctly and to gain full marks to parts (a) and (b). Part (c) required the equation for L for any marks to be scored. In general, only the better candidates could answer the final part correctly. Partial credit was awarded in part (a) to those who used embedded values but many of these candidates only applied the process to one of the required equations and often assumed the value of p to find the value of q only, then to use $q = 30$ to 'show' that $p = 0.6$, the value they had already used. Candidates who had any knowledge of this topic generally obtained the correct value for u_4 in part (b).